MCTA4363 DEEP LEARNING · TRACK 1 - KAGGLE COMPETITION

Development of an LLM-Based Intelligent Agent for the Kaggle LLM 20 Questions Competition

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Questioner Agent

Asks yes/no questions and makes final guesses

Answerer Agent

Responds strictly with "yes" or "no"

The Goal

Guess the hidden word in the fewest rounds possible

Project Objectives

- Develop a functional intelligent Questioner agent
- Compare rule-based vs. LLM-based approaches
- Evaluate reasoning performance quantitatively

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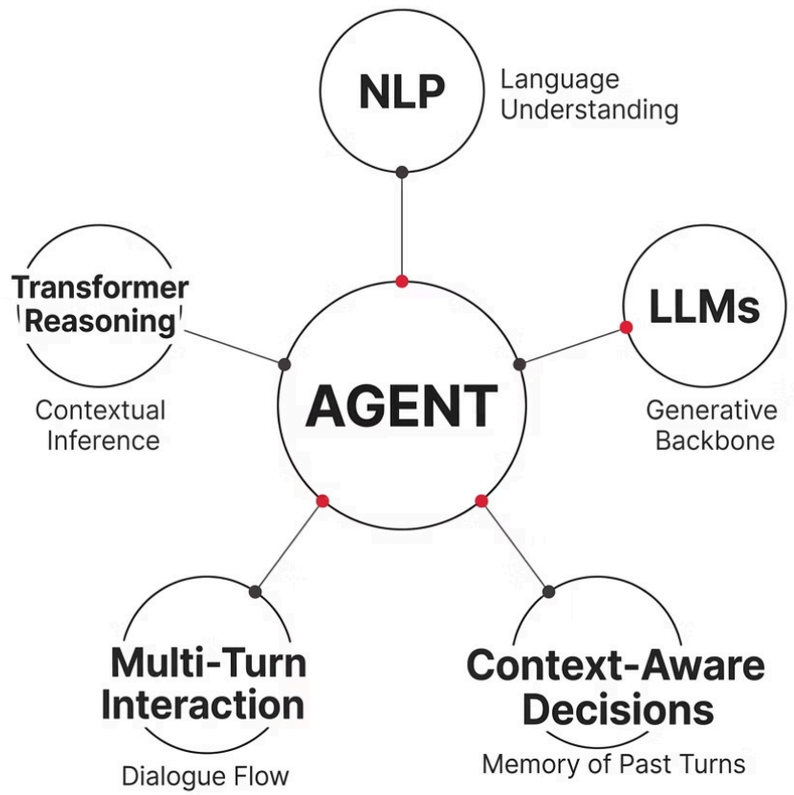
DEEP LEARNING RELEVANCE

Where Deep Learning Meets Strategic Reasoning



Core Technical Challenges

- **Generate meaningful questions** – avoid vague or redundant queries
- **Track conversation history** – maintain state across turns
- **Narrow possibilities logically** – binary search over concept space
- **Guess accurately under constraints** – maximize success in ⩽20 rounds

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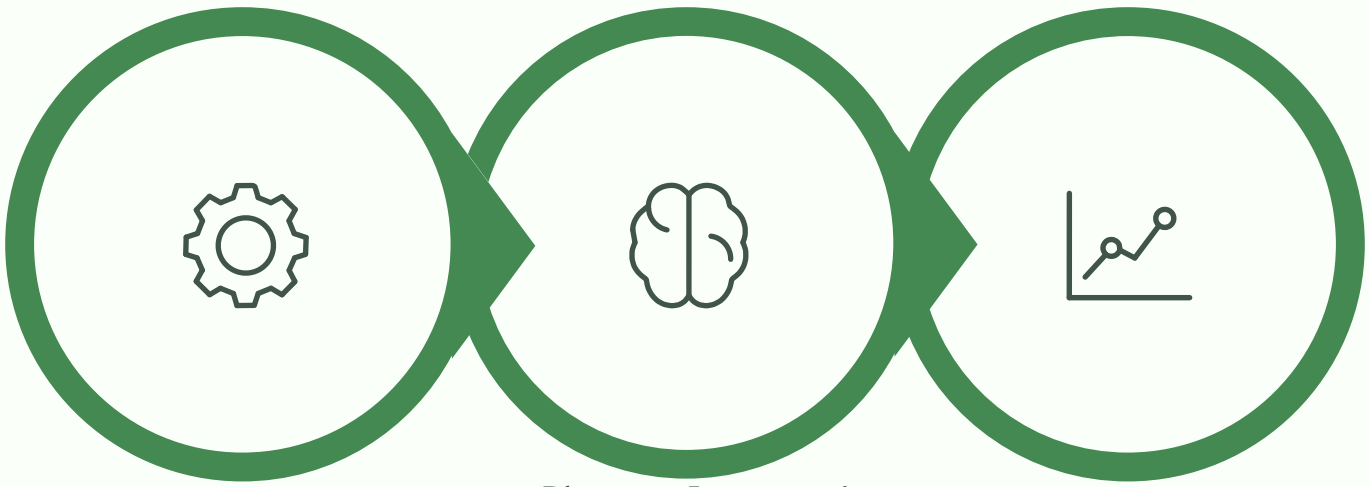
PROJECT PLAN

Objectives, Methodology & Expected Outcomes



Phase 1: Baseline System

-Rule Based question strategy

Phase 2: Improved LLM-Based Agent

-Use an LLM / prompting strategy to improve reasoning

Phase 3: Evaluation

The three-phase approach builds from a simple rule-based baseline to a full LLM-powered agent, with systematic evaluation at each stage.

Expected Outcomes

- A working intelligent agent for the Kaggle competition
- Better performance than a simple rule-based baseline
- Demonstration of LLM-based reasoning in an interactive setting

Expected Deliverables

Kaggle Submissions

GitHub Repository

Final Report

The LLM-based agent should significantly outperform the rule-based baseline in both success rate and average rounds, demonstrating the value of transformer-based reasoning in interactive tasks.

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